

Online Movie Ticket Booking System

Requirement Analysis

Introduction:

The Online Movie Ticket Booking System is designed to enable users to book movie tickets conveniently through an online platform. The system manages various aspects such as theaters, movie listings, shows, bookings, payments, and user reviews, while also providing alerts and generating reports.

Key Features:

- **User-friendly interface** for customers to browse movies and book tickets.
 - **Secure payment gateway** with discounts and refund handling.
 - **Theater management** for owners to update movie schedules and seat availability.
 - **Review and feedback system** to gather user opinions.
 - **Real-time notifications** for booking confirmations, cancellations, and upcoming shows.
 - **Comprehensive reporting** for analyzing ticket sales, popular movies, and theater performance.
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Entities and Attributes:

1. User:

- Attributes: userId, username, email, password, phone, role, address
- Role-based access (e.g., Customer, TheaterOwner, Admin)
- Actions: Register, login, update profile, view booking history

2. Movie:

- Attributes: movieId, title, genre, director, cast, releaseDate, duration, language, rating
- Actions: Add, edit, delete movie details (TheaterOwner/Admin only)

3. Theater:

- Attributes: theaterId, name, location, ownerId, totalScreens
- Actions: Add, edit, delete theater details (TheaterOwner/Admin only)

4. Show:

- Attributes: showId, movieId, theaterId, showTime, date, availableSeats, pricePerSeat
- Actions: Schedule shows for specific movies and theaters

5. Seat:

- Attributes: seatId, showId, seatNumber, status (Available, Booked, Blocked)
- Actions: View available seats, block/book seat

6. Booking:

- Attributes: bookingId, userId, showId, seatNumbers[], bookingStatus (Confirmed, Canceled), bookingDate, totalAmount

- Actions: Book, view, cancel booking

7. Payment:

- Attributes: paymentId, bookingId, userId, amount, paymentMethod (Credit Card, Debit Card, Net Banking, UPI), paymentStatus, transactionDate
- Actions: Process payments and refunds

8. Review:

- Attributes: reviewId, userId, movieId, rating, comments, reviewDate
- Actions: Submit reviews for movies

9. Notification:

- Attributes: notificationId, userId, message, notificationType (Booking Confirmation, Upcoming Show Reminder, Cancellation)
- Actions: Send notifications for different events

10. Invoice:

- Attributes: invoiceId, bookingId, userId, amount, invoiceDate
- Actions: Generate and view invoices for completed bookings

11. Feedback:

- Attributes: feedbackId, userId, feedbackContent, feedbackDate
- Actions: Submit feedback about the platform or service

12. Discount:

- Attributes: discountId, code, description, discountPercentage, validFrom, validTo, minimumPurchaseAmount
- Actions: Apply discounts during payment

13. TheaterOwner:

- Attributes: ownerId, theaterId, name, email, contact
- Actions: Manage theaters and movie schedules

14. Schedule:

- Attributes: scheduleId, theaterId, movieId, showTime, date
- Actions: Schedule movies for specific theaters

15. Role:

- Attributes: roleId, roleName (Admin, TheaterOwner, Customer)
- Actions: Manage roles and permissions for users

Modules and Functionality:

1. Movie and Theater Management:

- **Admin/TheaterOwner:**

- Add, edit, or delete movie and theater information.

- Manage show schedules, movie details, and assign showtimes.
- Monitor seat availability and update when needed.

- **User:**

- View movie details, check showtimes, and view available seats.

2. Booking System:

- **Customer:**

- Browse movie schedules and select preferred showtimes.
- Select available seats and proceed to booking.
- View and manage bookings (cancel if necessary).

- **System:**

- Check seat availability in real time.
- Automatically update seat status when booked or canceled.

3. Payment Gateway:

- **Customer:**

- Make payments for bookings using multiple methods (Credit/Debit Cards, Net Banking, UPI).
- Apply discount codes during checkout.
- View payment history and request refunds for canceled bookings.

- **System:**

- Handle transactions securely and generate invoices for successful payments.
- Manage refund processes for canceled bookings.

4. Review System:

- **Customer:**

- Submit reviews for movies after watching them.
- Provide ratings and comments about the theater's experience.

- **System:**

- Display reviews and ratings on movie details pages.
- Aggregate reviews to calculate the average rating for movies and theaters.

5. Notification System:

- **System:**

- Send notifications for upcoming shows, booking confirmations, and cancellations.
- Alert users of new movies added to their preferred theaters.
- Notify users about applied discounts or promotions.

6. Reporting Module:

- **Admin/TheaterOwner:**

- Generate reports on total ticket sales, movie popularity, and theater performance.
 - Analyze seat occupancy rates and trends for specific movies or times.
 - **System:**
 - Provide automated reports for business insights and decision-making.
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Technical Requirements:

Core Technologies:

- **Backend:** Core Java, Hibernate ORM for database interactions
- **Frontend:** JSP/Servlet or Spring MVC for user interface
- **Database:** MySQL/PostgreSQL to store and manage all entities
- **Payment Gateway:** Integration with third-party services (e.g., PayPal, Stripe)
- **Notifications:** Integration with email/SMS services for real-time alerts

Tools and Frameworks:

- **Hibernate (ORM Tool):** Manage object-relational mapping for all entities.
- **Collections API (Java):** Manage collections like lists for seat bookings, show schedules, etc.
- **JPA Annotations:** For mapping Java objects to database tables.
- **Maven/Gradle:** For project management and dependency handling.

Security Features:

- Role-based access control (Admin, Customer, TheaterOwner).
 - Encryption of sensitive user data (e.g., passwords, payment details).
 - Input validation and SQL injection prevention.
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Non-Functional Requirements:

- **Scalability:** The system should handle increasing users and theaters across multiple cities.
 - **Performance:** Ensure fast booking and payment processing with minimal delays.
 - **Reliability:** System should be available 24/7 without downtime.
 - **User Experience:** Simplified and intuitive interface for ease of use.
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This detailed requirement analysis forms the foundation for designing and developing the Online Movie Ticket Booking System, ensuring the system is feature-rich, scalable, and user-friendly.

Detailed Modules Description

The project consists of several key modules that define the core functionality of the system. Each module will handle specific aspects of the platform, from managing movies and theaters to booking tickets, processing payments, and sending notifications. Here's a detailed description of each module:

1. Movie and Theater Management Module

This module is responsible for managing all aspects related to movies, theaters, and their schedules.

Key Features:

- **Movie Management:** Admins and Theater Owners can add, update, or remove movie details such as title, genre, cast, director, release date, and duration.
- **Theater Management:** Theaters can be added, modified, or removed. Each theater will have multiple screens, and the seating arrangement for each screen can be defined.
- **Schedule Management:** Show schedules for different theaters can be created and managed. Admins or Theater Owners can assign movies to specific time slots in a theater.
- **Seat Availability:** The system will track seat availability in real-time. Once a seat is booked, it is marked as unavailable for others.
- **Showtimes and Schedules:** Customers can view movie showtimes for different theaters and filter by date, time, and movie preference.

Actions:

- Add, edit, and delete movies.
 - Add, edit, and delete theater information.
 - Schedule movie shows in different theaters.
 - View seating arrangements and seat availability for each show.
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2. Booking System Module

The Booking System Module allows users to book movie tickets, choose seats, and manage their bookings.

Key Features:

- **Seat Selection:** Customers can view available seats for a particular show in a theater and select their preferred seats for booking.
- **Booking Confirmation:** After selecting seats, customers can proceed to confirm their booking, making sure the selected seats are reserved for them.
- **View and Manage Bookings:** Users can view their past and current bookings. They can also cancel bookings if needed, as per the cancellation policy.
- **Seat Status Update:** Once a seat is booked, the system will update its status to "Booked" or "Unavailable" for others.

Actions:

- Browse and select movies and shows by date, time, and theater.
- View seating arrangements for each show and select seats.

- Confirm booking after seat selection.
 - Cancel bookings (if allowed by the system's policies).
 - View booking history and details.
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3. Payment Gateway Module

This module handles all the financial transactions related to booking, cancellations, and refunds.

Key Features:

- **Multiple Payment Methods:** Customers can pay for their bookings using a variety of payment methods such as Credit/Debit Cards, UPI, Net Banking, and Digital Wallets.
- **Discount Management:** During checkout, customers can apply discount codes, which will be validated and applied to the final payment.
- **Refund Management:** In the case of booking cancellations, the system can process refunds according to the booking and cancellation policies.
- **Invoice Generation:** Once the payment is successfully completed, the system generates an invoice for the transaction, which can be viewed by the user.

Actions:

- Secure processing of payments.
 - Validate and apply discount codes.
 - Generate and send invoices after successful payments.
 - Process refunds for canceled bookings.
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4. Review System Module

This module allows customers to share their reviews and ratings about the movies they've watched and the theaters they visited.

Key Features:

- **Submit Reviews and Ratings:** Customers can provide feedback for movies by giving a rating (1 to 5 stars) and writing comments about their experience.
- **View Reviews:** Other customers can view the reviews and ratings left by previous users to make informed decisions about which movies or theaters to choose.
- **Theater Reviews:** In addition to movie reviews, customers can rate and review the theaters they visited, commenting on aspects such as cleanliness, service, and seating comfort.

Actions:

- Submit reviews for movies after watching.
 - Rate theaters based on user experience.
 - View reviews and ratings by other users.
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5. Notification System Module

The Notification System Module keeps users updated about their bookings, upcoming shows, offers, and any changes related to the system.

Key Features:

- **Booking Confirmation:** Once a booking is successfully made, the system sends an immediate notification (via email/SMS) to confirm the ticket reservation.
- **Upcoming Show Reminders:** The system sends reminders to customers a few hours before their scheduled movie showtime to ensure they don't miss it.
- **Cancellation Alerts:** If a customer cancels a booking or a show gets canceled, notifications will be sent to inform the customer about the changes.
- **Promotions and Offers:** Users can be notified about special offers, discounts, or new movie releases via the notification system.

Actions:

- Send notifications for booking confirmations, cancellations, and show reminders.
 - Send alerts for upcoming movie releases or new promotions.
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6. Reporting Module

The Reporting Module is designed to provide business insights by generating reports based on various aspects of the movie booking system.

Key Features:

- **Ticket Sales Reports:** Generate reports on the total number of tickets sold for specific movies, theaters, or time periods.
- **Revenue Reports:** Analyze the total revenue generated from ticket sales, including details of payments, discounts applied, and refunds processed.
- **Popular Movies:** Generate reports that show which movies are the most popular among customers based on booking history and ratings.
- **Theater Performance:** Reports on how well each theater is performing in terms of occupancy rates, number of bookings, and customer reviews.

Actions:

- Generate ticket sales reports for movies, theaters, or specific time frames.
 - Create revenue reports with breakdowns of payments, discounts, and refunds.
 - Analyze popular movies and trends.
 - Evaluate the performance of theaters based on user reviews and occupancy rates.
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7. User Management Module

The User Management Module handles all user-related functionalities, including registration, authentication, and profile management.

Key Features:

- **User Registration:** New users can sign up by providing basic details such as name, email, phone number, and password. The system verifies the user and allows them to access the platform.

- **User Login and Authentication:** Users can log in using their credentials. Secure authentication ensures that only registered users can book tickets or perform other actions.
- **Role-Based Access Control:** Depending on the user's role (Customer, Admin, Theater Owner), access to different functionalities will be restricted accordingly. Admins have full control, Theater Owners can manage their theaters, and Customers can book tickets and manage their profiles.
- **Profile Management:** Users can update their personal information, such as email, phone number, and password.

Actions:

- User registration, login, and logout.
 - Role-based access to features (Admin, Theater Owner, Customer).
 - Update user profile information.
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8. Discount and Coupon Management Module

This module manages discount offers and coupons that users can apply to their bookings.

Key Features:

- **Create Discount Codes:** Admins can create discount codes with a specific percentage off or a fixed amount off the ticket price.
- **Discount Validation:** The system validates the discount codes entered by users to ensure they are still valid and applicable for the specific booking.
- **Apply Discounts:** Discounts are applied during checkout, reducing the total booking amount.
- **Expiry Management:** The system tracks the validity period of discount codes, automatically disabling expired codes.

Actions:

- Admins create and manage discount codes.
 - Validate and apply discounts during payment.
 - Track discount usage and effectiveness.
-

9. Feedback and Support Module

This module collects feedback from users and allows them to reach out to support for assistance.

Key Features:

- **Submit Feedback:** Users can provide general feedback about the platform, suggesting improvements or highlighting any issues they faced.
- **Customer Support:** Users can reach out to the support team for help with bookings, payments, or any other system-related issues.
- **Feedback Analysis:** The system collects feedback and helps the admin evaluate common issues or areas for improvement.

Actions:

- Submit platform feedback.

- Provide customer support and respond to user queries.
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10. Admin Dashboard Module

The Admin Dashboard provides an interface for the system administrator to manage the overall platform, including users, movies, theaters, and reports.

Key Features:

- **User Management:** Admins can view all users and manage their roles, block or unblock users, and reset passwords if needed.
- **Theater and Movie Management:** Admins can view and manage all theaters and movies listed on the platform.
- **Report Generation:** Admins can access reports generated by the reporting module for business insights.
- **Discount Management:** Admins can create and monitor discount campaigns and coupons.

Actions:

- Manage users, movies, theaters, and schedules.
 - Generate and view reports.
 - Manage discount campaigns.
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These modules together form a comprehensive system to manage online movie ticket bookings, providing both customers and administrators with the tools they need to make the process smooth, efficient, and secure.

Detailed Workflow

The **Online Movie Ticket Booking System** will involve a seamless flow of activities across multiple modules, ensuring smooth user interactions and system operations. Below is a detailed workflow covering various aspects, from user actions to backend processing and administrative tasks.

1. User Registration and Authentication Workflow

Actors:

- User (Customer, Theater Owner, Admin)
- System

Steps:

1. User Registration:

- A new user accesses the registration page.
- The user fills in details (Name, Email, Password, Phone Number, etc.).
- The system validates the input and creates a new account for the user.
- Confirmation email/SMS is sent for account verification.

2. Login:

- The user enters credentials (email/phone and password).
- The system validates the credentials.
- If valid, the user is granted access to their dashboard.
- Depending on the role (Admin, Theater Owner, Customer), the system redirects the user to the relevant dashboard.

3. Profile Management:

- The user can view and update their profile information (e.g., change email, phone number, or password).
- System updates the user's data in the database.

2. Movie and Theater Management Workflow (Admin/Theater Owner)

Actors:

- Admin/Theater Owner
- System

Steps:

1. Add Movie:

- Admin or Theater Owner accesses the movie management section.
- They enter the movie's details (Title, Genre, Cast, Duration, Release Date, Poster).
- The system validates and saves the movie details in the database.

2. Add Theater:

- Admin/Theater Owner enters theater details (Name, Location, Number of Screens, Seating Capacity for each screen).
- The system saves theater information in the database.

3. Create Show Schedule:

- Admin/Theater Owner selects a theater, movie, and screen.
- They select a date, time, and duration for the show.
- The system checks for conflicts (e.g., overlapping shows) and creates the schedule if valid.

4. View and Edit Schedules:

- Admin/Theater Owner can view all upcoming show schedules and modify them as needed (change time or movie).
- The system updates the schedule and notifies users if any changes affect existing bookings.

3. Booking Workflow

Actors:

- Customer
- System

Steps:

1. Browse Movies:

- The customer accesses the home page, where all available movies and theaters are listed.
- The user selects a movie to view details (Showtimes, Theaters, Ratings).

2. Select Theater and Showtime:

- The customer selects a theater and a preferred showtime.
- The system retrieves seat availability and displays a seat map for the selected show.

3. Seat Selection:

- The customer selects one or more available seats from the seat map.
- The system temporarily locks these seats (reserving them for a limited time) to prevent double bookings.

4. Proceed to Payment:

- The customer confirms the seat selection and proceeds to payment.
- The system calculates the total price (with applicable taxes and fees) and allows the customer to enter any discount codes.

5. Payment Processing:

- The customer selects a payment method (Credit/Debit card, UPI, Net Banking, Wallet).
- The system redirects to the secure payment gateway.

- Upon successful payment, the system confirms the booking and unlocks the reserved seats for the user.

6. **Booking Confirmation:**

- The system generates a booking confirmation, showing the details (Booking ID, Movie, Theater, Seat Number, Date/Time).
- An invoice is generated, and notifications are sent via email/SMS to the customer.

7. **Seat Status Update:**

- The system marks the booked seats as “Unavailable” in the seat map.

8. **View/Manage Bookings:**

- The customer can view all current and past bookings in their account.
- The customer can cancel a booking (if allowed by the theater policy). If canceled, the system processes a refund and updates seat availability.

4. **Payment and Invoice Workflow**

Actors:

- Customer
- System
- Payment Gateway

Steps:

1. **Payment Initialization:**

- After seat selection, the customer proceeds to payment.
- The system calculates the total price and applies any available discounts.

2. **Payment Gateway Redirection:**

- The customer selects a payment method.
- The system sends the payment request to the payment gateway (Credit/Debit card, UPI, Net Banking, etc.).
- The gateway processes the payment and returns the result (success/failure).

3. **Invoice Generation:**

- Upon successful payment, the system generates an invoice with payment details (Amount Paid, Transaction ID, Payment Date).
- The invoice is sent to the customer via email and is available for download from their profile.

4. **Refund Processing:**

- In case of cancellations, the system calculates the refund amount according to the policy.
- The refund request is sent to the payment gateway, and the customer is notified once the refund is processed.

5. **Review System Workflow**

Actors:

- Customer
- System

Steps:**1. Submit Movie/Show Review:**

- After attending a show, the customer can submit a review for the movie and the theater.
- The system prompts the customer to rate the movie (1-5 stars) and provide written feedback.

2. Submit Theater Review:

- The customer can also provide a separate rating for the theater based on cleanliness, service, and seating comfort.
- The review is saved in the database and made visible to other users.

3. View Reviews:

- Customers can view all existing reviews for a movie or theater before making a booking decision.
 - The system displays the average rating and individual reviews.
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6. Notification System Workflow**Actors:**

- System
- Customer

Steps:**1. Booking Confirmation:**

- After a successful booking, the system sends a confirmation notification to the customer (via email/SMS).

2. Upcoming Show Reminder:

- The system sends reminders to customers a few hours before their scheduled showtime.

3. Show Cancellation:

- If a scheduled show is canceled or rescheduled, the system notifies all customers with booked seats, informing them of the change.

4. Promotions and Offers:

- Customers receive notifications about ongoing promotions, discount codes, and upcoming movies.
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7. Reporting and Analytics Workflow**Actors:**

- Admin
- System

Steps:

1. Generate Ticket Sales Reports:

- Admins can select a date range and generate reports on ticket sales, broken down by movie, theater, or showtime.
- The system pulls data from the booking module and generates a visual report.

2. Revenue and Refund Reports:

- Admins can generate detailed reports on total revenue earned, discount usage, and refunds processed.

3. Movie and Theater Performance:

- Reports on the popularity of movies and the performance of theaters (number of bookings, average occupancy rate, user ratings) are generated.

4. Export and Save Reports:

- Admins can export reports in various formats (PDF, Excel, CSV) for further analysis or record-keeping.

8. Admin and Theater Owner Workflow

Actors:

- Admin
- Theater Owner
- System

Steps:

1. User Management (Admin):

- Admins can manage user accounts, including assigning roles, suspending accounts, or resetting passwords.

2. Theater and Movie Management (Admin/Theater Owner):

- Admins and Theater Owners can add, update, or delete movies, theaters, and show schedules.

3. Discount Management:

- Admins can create discount codes and promotional campaigns that apply to specific theaters, movies, or showtimes.

4. Performance Monitoring:

- Theater Owners can monitor the performance of their theaters (occupancy, reviews, revenue) using the reports generated by the system.

This detailed workflow outlines the step-by-step operations within the **Online Movie Ticket Booking System**, ensuring a smooth, user-friendly experience for customers while providing comprehensive management capabilities for admins and theater owners.